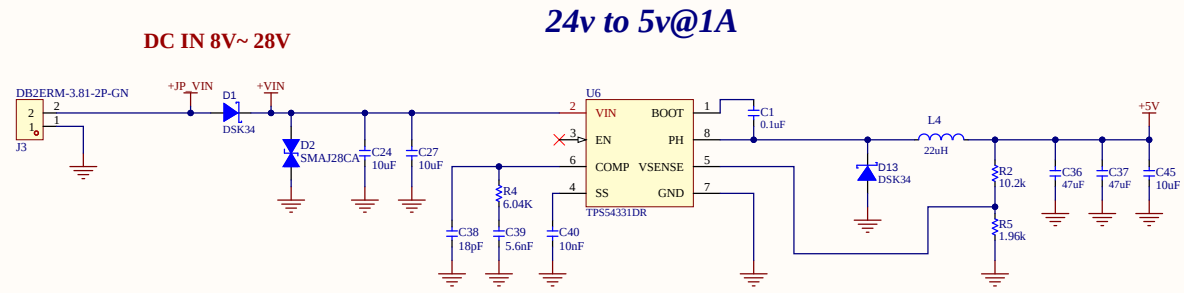
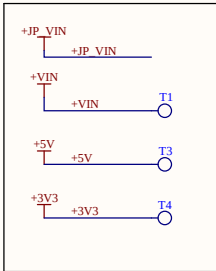
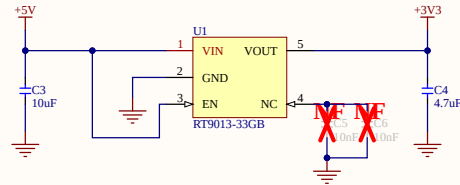


# 1 POWER

## Power.SchDoc



## 5v to 3v3@300mA



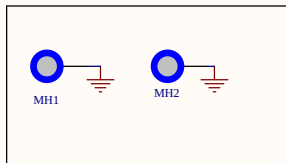
**Board Size: 70mm\*62mm**

Abbreviation Definition

**NF: Not Fitted**

**CK: Check the value of the components again**

## Mouting Holes



Logo EPCB

**MBW RF24 RS485 2.0**

Title: MBW\_RF24\_RS485\_210\_RevB.PrjPcb

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# 2 RS485

## RS485.SchDoc

RS485

PA3/RX2 >> PA3/RX2

PA1/RS\_DIR >> PA1/RS\_DIR

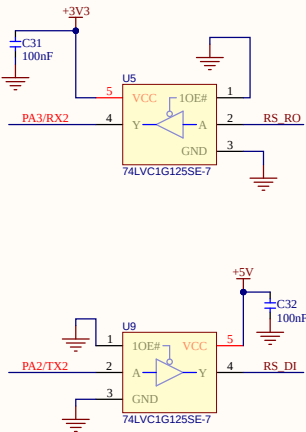
PA2/TX2 >> PA2/TX2

ID

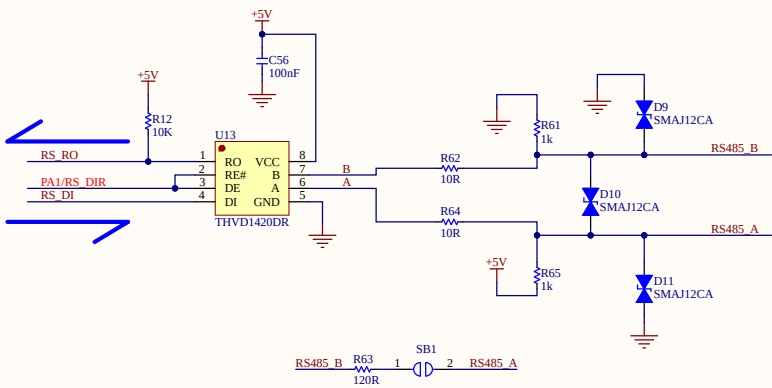
PC13/SW\_LOAD >> PC13/SW\_LOAD

PC14/SW\_SCK >> PC14/SW\_SCK

PC15/SW\_MISO >> PC15/SW\_MISO

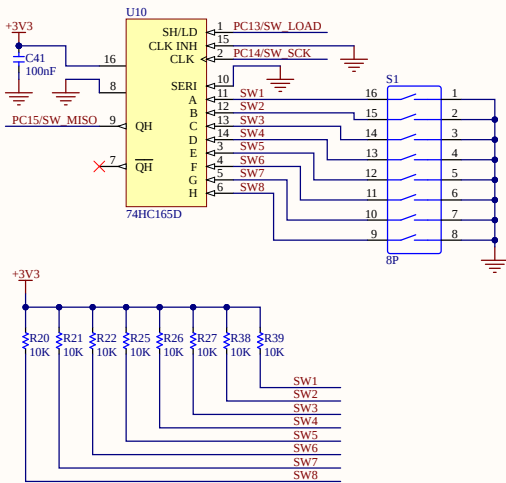


## RS485

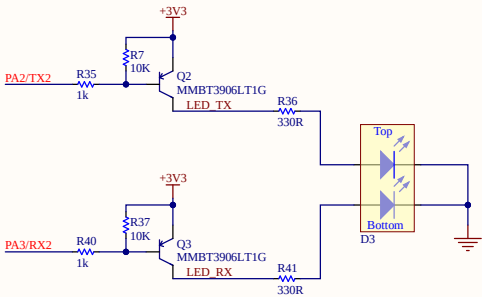


Solder bridge — open by default  
Short only when board is RS485 bus endpoint

## ID DIP Switches



## RS485 RX,TX LED Indicator

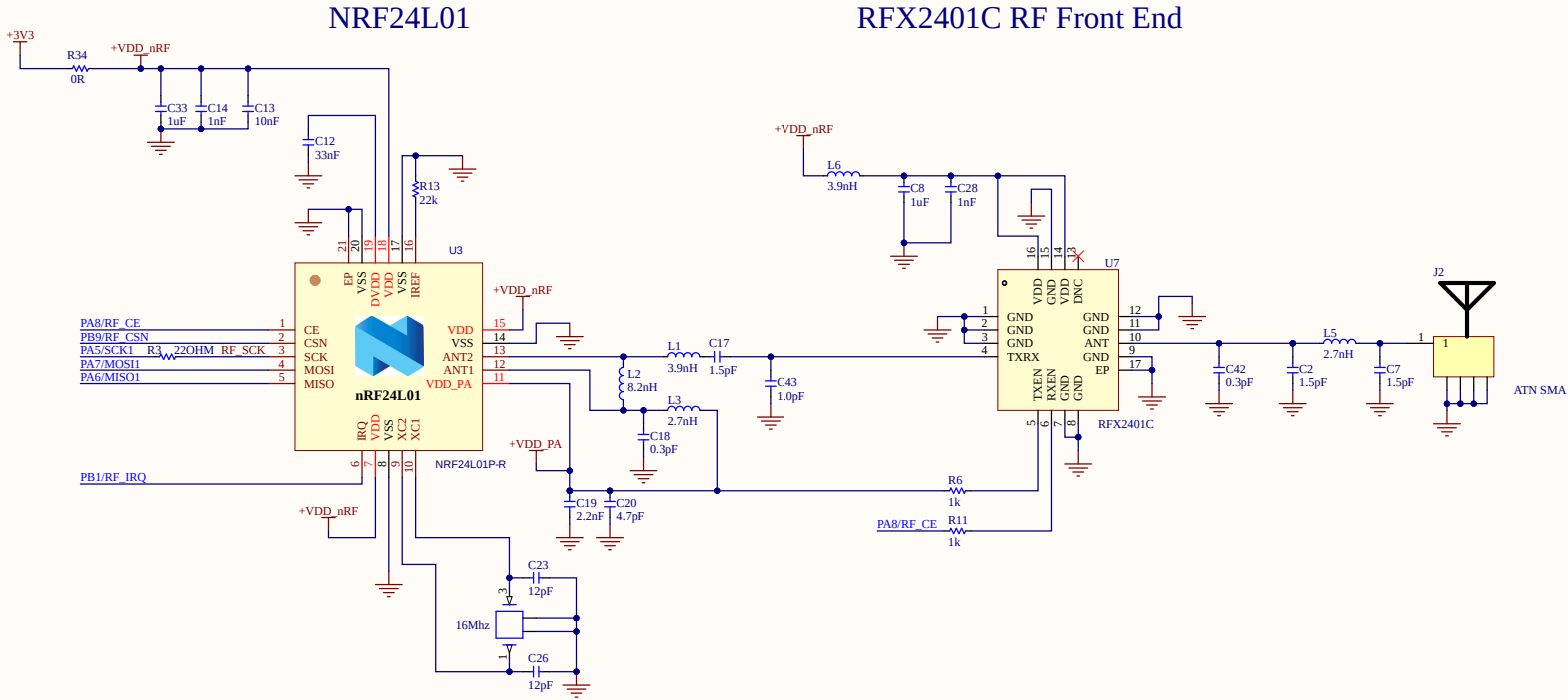
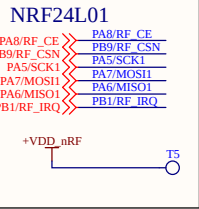


## MBW RF24 RS485 2.0

|                                       |                      |           |        |
|---------------------------------------|----------------------|-----------|--------|
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3 NRF24L01

NRF24L01.SchDoc



|                                       |              |             |  |
|---------------------------------------|--------------|-------------|--|
| MBW RF24 RS485 2.0                    |              |             |  |
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# 4 MCU

## MCU.SchDoc

### NRF24L01

PA8/RF\_CE PA8/RF\_CE  
PB9/RF\_CSN PB9/RF\_CSN  
PA5/SCK1 PA5/SCK1  
PA7/MOSI1 PA7/MOSI1  
PA6/MISO1 PA6/MISO1  
PB1/RF\_IRQ PB1/RF\_IRQ

### RS485

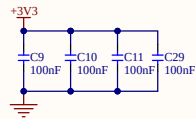
PA3/RX2 PA3/RX2  
PA1/RS\_DIR PA1/RS\_DIR  
PA2/TX2 PA2/TX2

### ID

PC13/SW\_LOAD PC13/SW\_LOAD  
PC14/SW\_SCK PC14/SW\_SCK  
PC15/SW\_MISO PC15/SW\_MISO

## STM32L151C8T6

## Expansion Board Connector



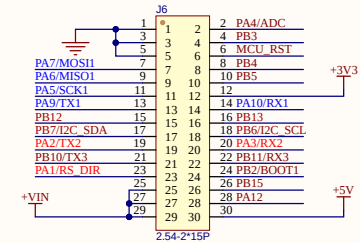
PA1/RS\_DIR  
PA2/TX2  
PA3/RX2  
PA4/ADC  
PA5/SCK1  
PA6/MISO1  
PA7/MOSI1  
PA8/RF\_CE  
PA9/TX1  
PA10/RX1  
PA11/BT\_RST  
PA12  
PA13/SWDIO  
PA14/SWCLK  
PA15/JTDI

PA0/USART2\_CTS/ADC12\_INT0/WKUP  
PA1/USART2\_RTS/ADC12\_IN1  
PA2/USART2\_TX/ADC12\_IN2  
PA3/USART2\_RX/ADC12\_IN3  
PA4/SPI1\_NSS/ADC12\_IN4  
PA5/SPI1\_SCK/ADC12\_IN5  
PA6/SPI1\_MISO/ADC12\_IN6  
PA7/SPI1\_MOSI/ADC12\_IN7  
PA8/MCO  
PA9/USART1\_TX  
PA10/USART1\_RX  
PA11/CAN\_RX/USNDP  
PA12/CAN\_TX/USBDP  
PA13/ITMS/SWDIO  
PA14/ITCK/SWCLK  
PA15/JTDI

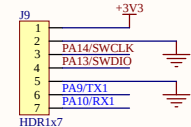
PB0/ADC12\_IN8  
PB1/ADC12\_IN9  
PB2/BOOT1  
PB3/JTDO  
PB4/JNTRST  
PB5  
PB6/I2C1\_SCL  
PB7/I2C1\_SDA  
PB8  
PB9  
PB10/USART3\_TX/I2C2\_SCL  
PB11/USART3\_RX/I2C2\_SDA  
PB12  
PB13  
PB14/W25CS  
PB15  
PC13/SW\_LOAD  
PC14/SW\_SCK  
PC15/SW\_MISO

PB0/BUZ  
PB1/RF\_IRQ  
PB2/BOOT1  
PB3  
PB4  
PB5  
PB6/I2C\_SCL  
PB7/I2C\_SDA  
PB8/LED\_LIFE  
PB9/RF\_CSN  
PB10/TX1  
PB11/RX1  
PB12  
PB13  
PB14/W25CS  
PB15  
PC13/SW\_LOAD  
PC14/SW\_SCK  
PC15/SW\_MISO

PB0/BUZ  
PB1/RF\_IRQ  
PB2/BOOT1  
PB3  
PB4  
PB5  
PB6/I2C\_SCL  
PB7/I2C\_SDA  
PB8/LED\_LIFE  
PB9/RF\_CSN  
PB10/TX1  
PB11/RX1  
PB12  
PB13  
PB14/W25CS  
PB15  
PC13/SW\_LOAD  
PC14/SW\_SCK  
PC15/SW\_MISO



## Debug

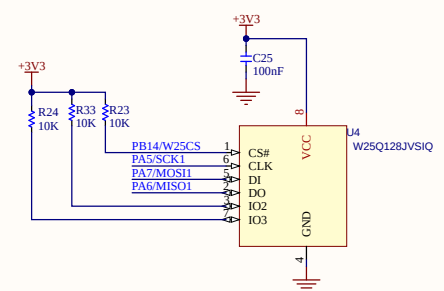
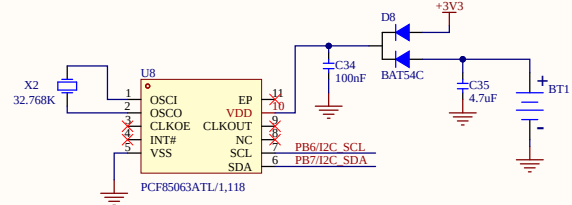
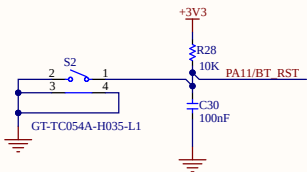
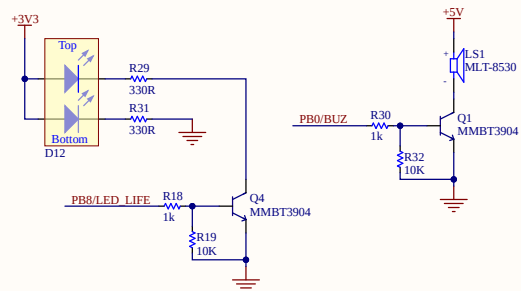


## Led & Buzzer

## User Reset

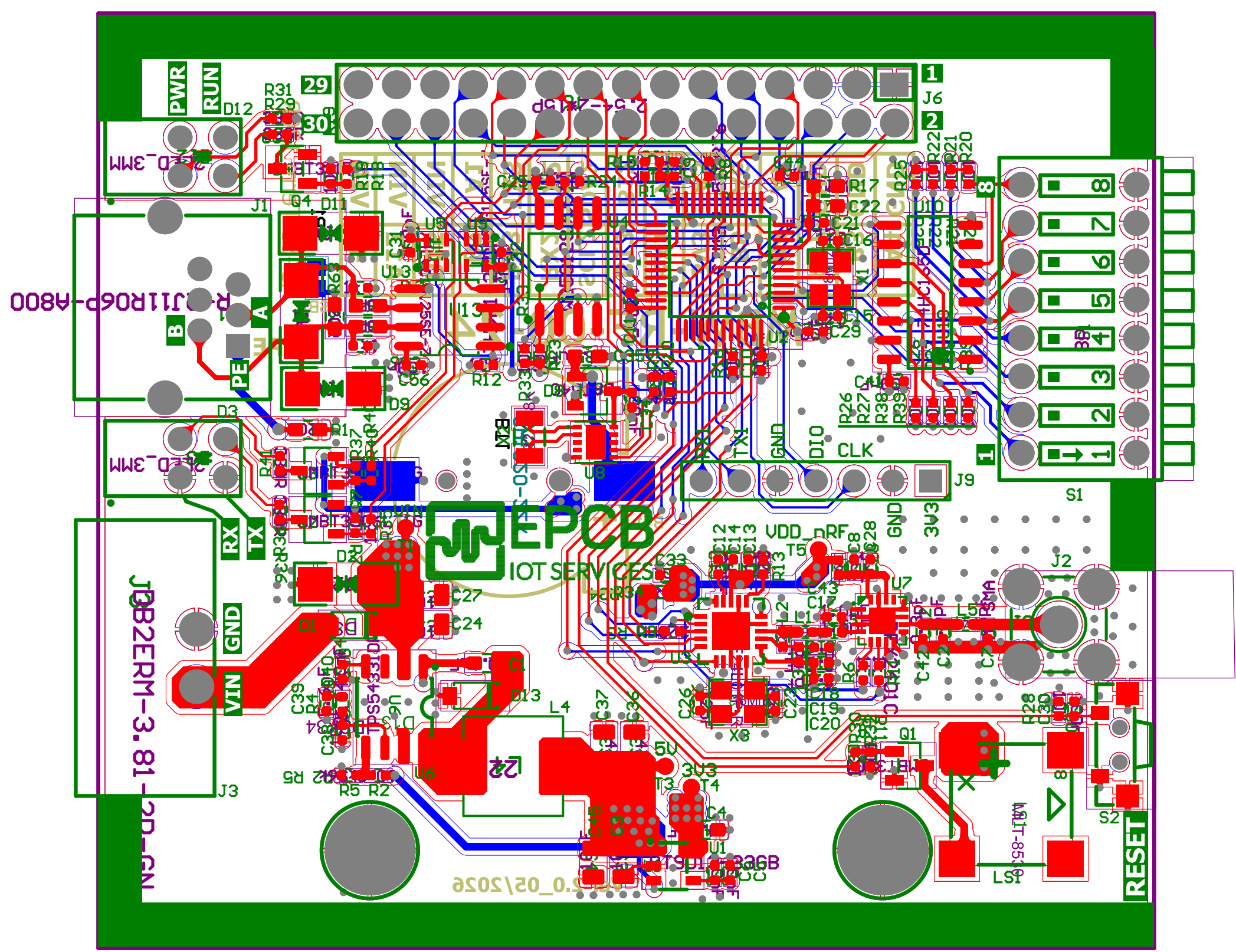
## RTC PCF85063

## NOR Flash



## MBW RF24 RS485 2.0

|                                       |           |              |  |
|---------------------------------------|-----------|--------------|--|
| Title: MBW_RF24_RS485_210_RevB.PrjPcb |           |              |  |
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**MBW\_RF24\_RS485\_2I0\_RevB.PrjPcb**

|    | Comment           | Designator | Footprint          | PCB Part Number | Manufacturer Part Number | Manufacturer                  | Quantity | Description            | Assembly Type  | Link |
|----|-------------------|------------|--------------------|-----------------|--------------------------|-------------------------------|----------|------------------------|----------------|------|
| 1  | CR1220-22X        | BT1        | BAT_CR1220_SMD     | C969906         | CR1220-22X               | Q&J                           | 1        | SMD Battery Holders, C | SMT Assembly   |      |
| 2  | 0.1uF             | C1         | C_0603             | C14663          | CC0603KRX7R9BB104        | YAGEO                         | 1        | 100NF 50V 0603 ROHS    | SMT Assembly   |      |
| 3  | 1.5pF             | C2, C7     | C_0402             | C76901          | GJM1555C1H1R5WB01D       | Murata Electronics            | 3        | 1.5pF 50V C0G 0402     | SMT Assembly   |      |
| 4  | 10uF              | C3, C24    | C_0805             | C15850, C4401   | CL21A106KAYNNNE, GRM     | Samsung Electro-Mechanics,    | 4        | 10UF 25V 0805 ROHS,    | SMT Assembly   |      |
| 5  | 4.7uF             | C4, C35    | C_0603             | C19666          | CL10A475K08NNNC          | Samsung Electro-Mechanics     | 2        | 4.7UF 16V 0603 ROHS    | SMT Assembly   |      |
| 6  | 1uF               | C8, C33    | C_0402             | C52923          | CL05A105KA5NQNC          | Samsung Electro-Mechanics     | 2        | 25V 1uF X5R ±10% 040   | SMT Assembly   |      |
| 7  | 100nF             | C9, C10    | C_0402             | C41851          | 0402B104K160NT           | FH(Guangdong Fenghua Adva     | 13       | 100nF 16V X7R ±10% 0   | SMT Assembly   |      |
| 8  | 33nF              | C12        | C_0402             | C106862         | CC0402KRX7R9BB333        | YAGEO                         | 1        | 50V 33nF X7R ±10% 04   | SMT Assembly   |      |
| 9  | 10nF              | C13, C4    | C_0402             | C15195          | CL05B103K85NNNC          | Samsung Electro-Mechanics     | 2        | 10nF 50V X7R ±10% 04   | SMT Assembly   |      |
| 10 | 1nF               | C14, C2    | C_0402             | C1523           | 0402B102K500NT           | FH(Guangdong Fenghua Adva     | 2        | 1nF 50V X7R ±10% 040   | SMT Assembly   |      |
| 11 | 12pF              | C15, C1    | C_0402             | C1547           | 0402CG120J500NT          | FH(Guangdong Fenghua Adva     | 4        | 12pF 50V C0G ±5% 040   | SMT Assembly   |      |
| 12 | 0.3pF             | C18, C4    | C_0402             | C85885          | GJM1555C1HR30BB01D       | Murata Electronics            | 2        | 0.3pF 50V C0G 0402     | SMT Assembly   |      |
| 13 | 2.2nF             | C19        | C_0402             | C1531           | 0402B222K500NT           | FH(Guangdong Fenghua Adva     | 1        | 2.2nF 50V C0G ±5% 04   | SMT Assembly   |      |
| 14 | 4.7pF             | C20        | C_0402             | C1569           | 0402CG47K7C500NT         | FH(Guangdong Fenghua Adva     | 1        | 4.7pF 50V C0G ±5% 04   | SMT Assembly   |      |
| 15 | 1uF               | C22        | C_0603             | C15849          | CL10A105KB8NNNC          | Samsung Electro-Mechanics     | 1        | 1UF 50V 0603 ROHS      | SMT Assembly   |      |
| 16 | 47uF              | C36, C3    | C_0805             | C124129         | GRM21BR61A476ME15L       | Murata Electronics            | 2        | 47uF ±20% 10V X5R 08   | SMT Assembly   |      |
| 17 | 18pF              | C38        | C_0402             | C1549           | 0402CG180J500NT          | FH(Guangdong Fenghua Adva     | 1        | 50V 18pF C0G ±5% 040   | SMT Assembly   |      |
| 18 | 5.6nF             | C39        | C_0402             | C1540           | 0402B562K500NT           | FH(Guangdong Fenghua Adva     | 1        | 5.6nF 50V X7R ±10% 0   | SMT Assembly   |      |
| 19 | 1.0pF             | C43        | C_0402             | C76953          | GRM1555C1H1R0CA01D       | Murata Electronics            | 1        | 1.0pF 50V C0G 0402     | SMT Assembly   |      |
| 20 | DSK34             | D1, D13    | SOD-123_L2.8-W1.8  | C41029          | DSK34                    | MDD (Microdiode Electronics   | 2        | DSK34 40V 550mV@3      | SMT Assembly   |      |
| 21 | SMAJ28CA          | D2         | SMA_DO-214AC - C   | C113964         | SMAJ28CA                 | MDD(Microdiode Semiconduct    | 1        | -65°C~+150°C 1uA 28V   | SMT Assembly   |      |
| 22 | 2LED_3MM          | D3, D12    | 2LED_3MM           | N/A             | SSF-LXH240IGD            | Lumex                         | 2        | LED DUAL RED/GREE      | Hand Soldering |      |
| 23 | BAT54C            | D8         | SOT-23-3_L2.9-W1   | C14308          | BAT54CLT1G               | ON Semicon                    | 1        | SCHOTTKY BARRIER       | SMT Assembly   |      |
| 24 | SMAJ12CA          | D9, D10    | SMA_DO-214AC - C   | C698919         | SMAJ12CA                 | Yangzhou Yangjie Elec Tech    | 3        | 1uA 20.1A 19.9V 13.3V  | SMT Assembly   |      |
| 25 | R-RJ11R06P-A800   | J1         | RJ11-TH_PCB-6P60   | C189750         | R-RJ11R06P-A800          | Ckmtw(灿科盟)                    | 1        | R-RJ11R06P 0°C~+70°    | Hand Solder    |      |
| 26 | ATN SMA           | J2         | ANTEN_SMA          | N/A             | N/A                      | N/A                           | 1        | Connector; SMB Straig  | Hand Soldering |      |
| 27 | DB2ERM-3.81-2P-G  | J3         | CONN-TH_DB2ERM     | C430325         | DB2ERM-3.81-2P-GN        | DORABO(地博电气)                  | 1        | -40°C~+105°C 1 1x2P 2  | SMT Assembly   |      |
| 28 | 2.54-2*15P        | J6         | HDR-TH_30P-P2.54   | C5337168        | 2.54-2*15P               | BOOMELE(博穆精密)                 | 1        | Straight,P=2.54mm Fe   | SMT Assembly   |      |
| 29 | 3.9nH             | L1, L6     | L_0402             | C98063          | LQW15AN3N9C00D           | Murata Electronics            | 2        | 3.9nH 70mΩ 750mA Ur    | SMT Assembly   |      |
| 30 | 8.2nH             | L2         | L_0402             | C317946         | FWW0402UC8N2JST          | FH(Guangdong Fenghua Adva     | 1        | 8.2nH 680mA ±5% 100    | SMT Assembly   |      |
| 31 | 2.7nH             | L3, L5     | L_0402             | C27123          | SDCL1005C2N75TDF         | Sunlord                       | 2        | 2.7nH 200mΩ 6GHz 80    | SMT Assembly   |      |
| 32 | 22uH              | L4         | IND-SMD_L7.1-W6.6  | C207843         | SLO0630H220MTT           | Suntech Tech                  | 1        | 22uH ±20% SMD Powe     | SMT Assembly   |      |
| 33 | MLT-8530          | LS1        | BUZ-SMD_4P-L8.5-W  | C94599          | MLT-8530                 | Jiangsu Huaneng Elec          | 1        | BUZZERS MAGNETIC       | SMT Assembly   |      |
| 34 | MMBT3904          | Q1, Q4     | SOT-23-3_L2.9-W1   | C20526          | MMBT3904                 | Changjiang Electronics Tech ( | 2        | MMBT3904 TRANSIST      | SMT Assembly   |      |
| 35 | MMBT3906LT1G      | Q2, Q3     | SOT-23-3_L2.9-W1   | C53444          | MMBT3906LT1G             | ON Semicon                    | 2        | MMBT3906 TRANSIST      | SMT Assembly   |      |
| 36 | 120R              | R1, R63    | R_0603             | C22787          | 0603WAF1200T5E           | UNI-ROYAL(Uniroyal Elec)      | 2        | 120 OHMS ±1% 1/10W     | SMT Assembly   |      |
| 37 | 10.2k             | R2         | R_0402             | C226746         | AC0402FR-0710K2L         | YAGEO                         | 1        | 10.2k ±1% 10.2kΩ 0402  | SMT Assembly   |      |
| 38 | 220HM             | R3         | R_0402             | C25092          | 0402WGF220JTCE           | Uniroyal Elec                 | 1        | 22Ω 0402 62.5mW Thic   | SMT Assembly   |      |
| 39 | 6.04K             | R4         | R_0402             | C97819          | RC0402FR-076K04L         | YAGEO                         | 1        | 6.04K ±1% 6.04kΩ 040   | SMT Assembly   |      |
| 40 | 1.96k             | R5         | R_0402             | C324853         | RC-02K1961FT             | FH(Guangdong Fenghua Adva     | 1        | 1.96k ±1% 1.96kΩ 0402  | SMT Assembly   |      |
| 41 | 1k                | R6, R11    | R_0402             | C11702          | 0402WGF1001TCE           | Uniroyal Elec                 | 8        | 1kΩ 0402 62.5mW Thic   | SMT Assembly   |      |
| 42 | 10K               | R7, R8, R  | R_0402             | C25744          | 0402WGF1002TCE           | Uniroyal Elec                 | 19       | 10kΩ 0402 62.5mW Th    | SMT Assembly   |      |
| 43 | 4.7k              | R9, R10    | R_0402             | C25900          | 0402WGF4701TCE           | Uniroyal Elec                 | 4        | 4.7kΩ 0402 62.5mW Th   | SMT Assembly   |      |
| 44 | 22k               | R13        | R_0402             | C25768          | 0402WGF2202TCE           | Uniroyal Elec                 | 1        | 22k 0402 62.5mW Thic   | SMT Assembly   |      |
| 45 | 00HM              | R17        | R_0603             | C21189          | 0603WAF0000T5E           | Uniroyal Elec                 | 1        | 00HMS ±1% 1/10W 06     | SMT Assembly   |      |
| 46 | 330R              | R29, R3    | R_0402             | C12246          | 0402WGJ0331TCE           | Uniroyal Elec                 | 4        | 330R ±5% 330Ω 0402     | SMT Assembly   |      |
| 47 | 0R                | R34        | R_0402             | C17168          | 0402WGF0000TCE           | Uniroyal Elec                 | 1        | 0R 0Ω 0402 1% Chip R   | SMT Assembly   |      |
| 48 | 10R               | R62, R6    | R_0603             | C22859          | 0603WAF100JT5E           | UNI-ROYAL(Uniroyal Elec)      | 2        | 100mW Thick Film Res   | SMT Assembly   |      |
| 49 | 8P                | S1         | SW-TH_KF1002-08P   | N/A             | KF1002-08P-R3GLG-DFT     | KAIFENG                       | 1        | Plugin DIP Switches R  | Wave Soldering |      |
| 50 | GT-TC054A-H035-L  | S2         | SW-SMD_L7.8-W3.4   | C778158         | GT-TC054A-H035-L1        | G-Switch(品赞)                  | 1        | None Without 50mA 7.8  | SMT Assembly   |      |
| 51 | RT9013-33GB       | U1         | SOT-23-5_L3.0-W1   | C47773          | RT9013-33GB              | Richtek Tech                  | 1        | 5.5V 300mV @ 500mA     | SMT Assembly   |      |
| 52 | STM32L151C8T6     | U2         | LQFP-48_L7.0-W7.0  | C9861           | STM32L151C8T6            | STMicroelectronics            | 1        | STM32 ARM-based 32-    | SMT Assembly   |      |
| 53 | NRF24L01P-R       | U3         | QFN-20_L4.0-W4.0   | C8791           | NRF24L01P-R              | NORDIC                        | 1        | NRF24L01P-R SPI QFN    | SMT Assembly   |      |
| 54 | W25Q128JVSQ       | U4         | SOIC-8_L5.3-W5.3   | C97521          | W25Q128JVSQ              | Winbond Elec                  | 1        | -40°C~+85°C 100,000 c  | SMT Assembly   |      |
| 55 | 74LVC1G125SE-7    | U5, U9     | SOT-353_L2.1-W1.3  | C151022         | 74LVC1G125SE-7           | Diodes Incorporated           | 2        | 74LVC1G125SE-732m      | SMT Assembly   |      |
| 56 | TPS54331DR        | U6         | SOIC-8_3.9x4.9x1.2 | C9865           | TPS54331DR               | Texas Instruments             | 1        | TPS54331DR 3A 28V S    | SMT Assembly   |      |
| 57 | RFX2401C          | U7         | QFN-16_L3.0-W3.0   | C19213          | RFX2401C                 | SKYWORKS                      | 1        | ERX2401C 2.4GHz~2.5    | SMT Assembly   |      |
| 58 | PCF85063ATL/1,118 | U8         | DFN-10_L2.6-W2.6   | C404360         | PCF85063ATL/1,118        | NXP(恩智浦)                      | 1        | PCF85063ATL/1,118 0    | SMT Assembly   |      |
| 59 | 74HC165D          | U10        | SOIC-16_3.9x9.9x1  | C5613           | 74HC165D,653             | Nexperia                      | 1        | 74 SERIES SHIFT REG    | SMT Assembly   |      |
| 60 | THVD1420DR        | U13        | SOIC-8             | C3235215        | THVD1420DR               | Texas Instruments             | 1        | transceiver 4.75V~5.25 | SMT Assembly   |      |
| 61 | 8Mhz              | X1         | SMD-3225_4P        | C133366         | 1C208000CE0Q             | KDS Daishinku                 | 1        | 8Mhz 3225 SMD CRYST    | SMT Assembly   |      |
| 62 | 32.768K           | X2         | SMD-3215_2P        | C390740         | SMD31327681252090        | JGHC                          | 1        | 32.768K -40°C~+85°C    | SMT Assembly   |      |
| 63 | 16Mhz             | X3         | SMD-3225_4P        | C13738          | X322516MLB4SI            | Yangxing Tech                 | 1        | 16MHz SMD Crystal Re   | SMT Assembly   |      |